CHAPTER 4

Data design

This chapter will focus on answering five key questions:

- Why collect multiple types of data?
- Which collection methods should you combine?
- What are the possibilities and limitations of different types of data?
- How can different data and data collections be integrated?
- How do you assess data quality?

First, we will look at how combinations of collection methods and data types can ensure greater creativity, quality and efficiency in a study. We then return to the division into research styles presented in Chapter 2 to show how methodological perspectives and combinations between research styles and scale levels can be considered when developing an actual data design. We also review basic distinctions of data properties in relation to capturing the social world and how these differences can be helpful when choosing data types for your study. We then look at different forms of data integration and data transformation and how these can be applied in data design. At the end of the chapter, we turn to the key issues regarding *the quality* of mixed methods data design. A crucial prerequisite for different data types to complement each other is that they have the necessary quality. Therefore, we review the key data quality criteria that are typically relevant in mixed methods projects.

Mixed methods studies allow you to know more and be able to say more about a phenomenon or problem compared to single method studies. The reason why mixed

methods studies can do this is because different types of data and data collection provide different access to reality, and combinations of data types can therefore create more comprehensive, in-depth, precise or versatile knowledge about a phenomenon. In chapters 2 and 3, we touched on the fact that method specialisation has led many social scientists to use only one type of data, and that all their research questions and creative ideas must find their answers through that one data type. Economists, for example, work almost exclusively with found large-scale data, while anthropologists and ethnographers swear by observations through fieldwork and smallscale data. As a result, they ask research questions that fit the particular data types they are used to using, and often avoid questions that cannot be answered with that type of data. This leads to baroque cases where different studies from different disciplines that are actually studying the same phenomenon seem to come from different planets and where the phenomenon, besides its name, can be difficult to recognise across studies.

Data collection and data sources are the 'keyhole' that the researcher can look through to look at reality, and two different keyholes can yield very different observations. Always using the same keyhole no matter what you're investigating creates an impoverished social knowledge creation with an atomised and disjointed understanding of reality. It also creates knowledge that is difficult to put into practice and solve real-world problems because such problems exist across scales, data types and methodological perspectives. It's only when we combine the above that we get closer to a comprehensive understanding of the problems.

A mixed methods perspective can help us unlock a wealth of diverse data that can be combined to better understand social reality. By wisely selecting different data gaps that need to be filled, we get several different perspectives on reality that can be analysed and interpreted to create a better and more comprehensive picture of reality. Therefore, a crucial part of developing a mixed methods design is to consider how best to capture and elucidate the phenomenon through data collection, what characteristics of data you need, and how best to combine them to answer your research questions. That is the topic of this chapter.

Why collect multiple types of data: Creativity, quality and efficiency

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In 1967, Glaser and Strauss encouraged the use of a wide range of data, especially for theory development: "Sociologists can greatly expand the range of qualitative data that can be used to generate theory with relatively little expenditure of time, money and effort" (Glaser & Strauss, 1967: 161). Glaser and Strauss' statement emphasises both the analytical potential of using multiple types of data, but also that it can be a resource-efficient approach to research. The Marienthal study, which we presented in chapter 2, is again a good source of inspiration. The many different consequences of unemployment were uncovered by combining a multitude of different data types that complemented each other in different ways. The combination of e.g. family diaries and indepth interviews with family members is an obvious way to generate and refine theories about the impact of unemployment on individuals and their close relationships. Family diaries can be used to discover consequences and problems that researchers had not previously considered. This allows the researcher to start developing new theories about unemployment. The in-depth interviews can then be used to deepen, supplement and test the researchers' interpretations of the knowledge gained from the diaries. In other words, they are complementary data sources.

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For Glaser and Strauss, the most important thing is that we use the data that best facilitates ongoing theory development, so we avoid our ideas and wonder being limited by a predetermined data base. In other words, we should let the choice of data types and data combinations be driven by the ongoing development of ideas about what we are investigating. Writing in 1967, Glaser and Strauss focused on archival and library data as an underutilised source (Abbott, 2014) and were particularly critical of our lack of creativity when it comes to finding and collecting data. Since then, social reality has only become increasingly full of data sources, and we need to develop a 'datafication' view of reality in order to utilise the potential of these many new data sources.

The flexible use of multiple data types not only helps to develop theories, but ultimately typically provides a better understanding of social reality and

supports higher quality conclusions with greater validity and reliability. This is because the processes we typically want to understand require that we both gain insight into people's own interpretation of their situation, their actions and their interactions with others, and that we understand the institutional framework and structural conditions that prevail in the situation. All these aspects cannot be captured with just one type of data, but require data that can capture subjective interpretation, situated actions, interaction, institutional conditions, structures and the historical context. At the same time, we also want data that enables the results to be generalised beyond the specific case or cases we are studying.

These are claims that data from interviews, observations, experiments or found data alone cannot fulfil. Therefore, the quality of results based on only one type of data can be jeopardised by focusing on the blind spots of this data: the perspectives that the data source cannot capture. This is one of the problems that can be addressed by using multiple methods and data types in a mixed methods data design.

Another argument in favour of using multiple data types is one of efficiency: How quickly and cheaply can we obtain acceptable evidence for our statements? All studies are limited in terms of the amount of data that can be collected and analysed, and you have to prioritise your resources: where do we best spend our time and material resources to answer the problem statement? When we are unsure about something in our research, what should we do? Should we do more interviews? Ask more survey questions? The most effective way to obtain sufficient evidence for a theory is often to draw on multiple data types simultaneously, rather than collecting more of the data type you've already collected. For example, if we have found a very strong correlation in a statistical analysis, but cannot clearly find an explanation in the data as to why this correlation occurs, then it is not efficient to collect more data of the same type. Instead, it may be possible to determine which of several possible micro-level processes is creating this correlation by conducting targeted, in-depth interviews or ethnographic observations. Such combinations will often effectively address questions about underlying mechanisms, and at the same time, the complementary data can deepen the understanding of the phenomenon and explore dimensions or relationships that large-scale data alone did not provide access (see example in text box 4.1).

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Box 4.1 Mixed methods effectiveness: The relationship between trust in political institutions and activism

Data design

An example of effective use of different data types is Jonas Toubøl's study "From Democratic Participation to Civic Resistance: The Loss of Institutional Trust as an Outcome of Activism in the Refugee Solidarity Movement" from 2019. It studies the relationship between institutional trust and activism. The common explanation in the literature on institutional trust is that low trust in

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political institutions (e.g. parliament, police, state administration, judiciary, etc.) leads to activism: when you don't trust politicians and state institutions, you will choose activism and take to the streets to protest against them because you don't trust that your views will be heard through the institutionalised channels. The problem was that this conclusion was based on cross-sectional survey data, which did not allow us to analyse whether the causal link between distrust and activism went one way or the other. Another possible explanation was that perceptions of fair or unfair treatment from political institutions in connection with activism influenced whether individuals' trust in institutions increased or decreased. If you experienced that the police were unfair at an otherwise peaceful demonstration, you will have less trust in the police in the future. The problem with activism and protest is that it's hard to predict and therefore difficult to ask people about their trust in institutions before they engage in activism. Therefore, large-scale data comparing individuals before and after they engage in activism is virtually non-existent. This was also the situation Toubøl was in. It would be extremely expensive to collect panel survey data that could measure trust before and after activism in a representative sample. Instead, Toubøl used in-depth, life-history interviews and explored how respondents' relationship with institutions was affected by their activism. These interviews provided significant evidence in favour of the hypothesis that encountering institutions through activism could lead to a loss of trust, and that the loss of trust was linked specifically to the institutions one had interacted with through activism. This was also indicated by the cross-sectional surveys used, which asked whether participation in different forms of activism had changed participants' trust in different political institutions.

Thus, the two data types, in-depth interviews and surveys, complemented each other and allowed for a generalisation of the results of the interviews.

Choosing data collection methods

Mixed methods research can provide better or more knowledge and more efficient use of resources if you combine the right types of data. But what are the right data collection methods? Once you have a problem statement and perhaps some research questions in place, a key part of the task in the design phase is to ask yourself: How and where can I capture the phenomenon I'm interested in? How should I gather data about the phenomenon to answer my research question?

The first requirement for choosing a data collection method is the same in single and mixed methods research: it must produce data in a way that is suitable for capturing the phenomenon you want to study. Video recordings of polling stations are not suitable for capturing political beliefs, semi-structured interviews are not suitable for measuring the extent of criminal activity in a population, and it's difficult to do fieldwork in a way that allows you to observe people's thoughts. The method must lend itself to studying the phenomenon, or to put it another way: you need to choose a data keyhole that faces into the space you want to look into. In social science research, phenomena are very much something that needs to be defined and operationalised before you can study them. Therefore, choosing the right data collection method is both a question of how reality is constructed and how you theoretically define and operationalise what you are looking at.

For example, if you are interested in how and when people trust each other, you can define trust in different ways that lend themselves to study with different data collection methods. If trust is defined as the willingness to be vulnerable with other people, you may need to do observational studies of people in situations where they can choose to make themselves vulnerable or not. For example, situations where they are dependent on the help and advice of others or where they have to leave something important, such as their children, in the care of others.

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If, on the other hand, trust is defined as a kind of expectation of other people - an expectation that others will act trustworthy - we need to talk to them to capture their subjectively perceived expectations. If trust is a kind of risk-taking, you can design some game-like experiments to find out who trusts others and how much risk they are

willing to take based on that trust. Other definitions will lead to other methodological choices being possible and right.

This is nothing new compared to single-method research, as long as you don't just adapt the definition of the phenomenon to the methods you are used to. The key is that it is the problem statement's definition of the phenomenon you are interested in studying that determines which data and methods you use to collect it. Table of contents 4.1 provides an overview of which aspects of social reality the research styles presented in chapter 2 are particularly suited to capture.

Table 4.1 What phenomena do the research styles capture?

Research style			
Interview	*Observation*	*Found data*	*Experimentation*
Phenomena that people have a conscious understanding of and can talk about Phenomena that exist as systems and patterns in human speech and stories	Phenomena that have a location (physical or virtual) where actions and stories take place in a social practice that can be seen, heard and possibly even participated in	Phenomena that leave interpretable traces as images, numbers or text, either physically or virtually Phenomena found as systematics and patterns in the traces left behind	Phenomena that can be operationalised in interactions, influences or choice situations with clear action alternatives

In Chapter 2, we showed how the distinction between quantitative and qualitative data and methods is only useful in a very limited way. It is especially useless when it comes to the task of choosing data and collection methods. It's about considering which data and collection methods are best suited to capture a phenomenon in the specific way you have defined it. Among the many logics that can be used to select data collection methods in a mixed methods study, we will emphasise two in particular.

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The first is the logic of *exclusion* we touched on above, where you consider whether there is any data that is not useful for capturing the phenomenon you are investigating. In some cases, this may leave a range of possible data and collection methods that are useful, and in other situations, there may only be a few types of data that can be considered.

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The second is the *logic of complementarity*, which is a direct extension of the complementarity considerations we discussed in chapter 3 on design. You may have an understanding of the research phenomenon that includes both an action aspect and an attitude aspect. Then it makes sense to either use some kind of fieldwork or to combine interview and observation. In the example of trust, this could mean, for example, studying trusting actions through observations and people's perception of what makes others trustworthy through interviews. It can also be crucial whether the theory you have about the phenomenon is one that needs to be tested or whether the intention is to develop theory. Experiments and closed survey interviews are particularly suitable for theory testing, while fieldwork and open or in-depth interviews are particularly suitable for theory development, without saying that you cannot develop theory with surveys or, conversely, test theories with fieldwork and in-depth interviews.

The choice of research style and the choice of specific data collection methods within each research style is primarily dependent on three factors
factors: 1) How you have defined the purpose of the study and the phenomenon you are investigating, 2) what you can access to collect data on, and 3) how you want the methods to complement each other (see chapters 3 and 5).

However, there are many other considerations that can come into play when choosing collection methods and data types. How important is it to avoid influencing the reality you are analysing? In some situations it doesn't matter so much, in other situations where people are inclined to hide or change their attitudes or actions, it is very important to think about the impact of data collection on the phenomenon you want to capture in your data. If you're trying to change people's behaviour, it can be important to understand them in their 'natural' social context, for example through observational studies. However, if you are testing a theoretical hypothesis, it can be

advantageous not to have a confusing and unpredictable context influencing actions simultaneously, which is why artificial, experimental data collection is favoured.

Table 4.2 Researcher impact and natural/artificial data production within the research styles

	Research style			
	Interview	*Observation*	*Found data*	*Experimentation*
Natural/ Not scientist influenced		Observational data will in most cases be of naturally occurring interactions and actions The presence of an observing researcher, who may also ask questions, can create an artificial element in data production	Data that is not created by or for the researcher or not perceived by participants as data production is always natural and without researcher influence	Experiments are only natural if they are naturally occurring or are field experiments
Artificial/ Scientist Influenced	Interviews are always staged. It is an artificial data production, often involving researcher influence from an interviewer			Most experiments are artificially staged and participants are aware of it. Often there will be a researcher present who influences the participants

In other words, there are a number of differences that we usually wrap up in the qualitative/quantitative division that are relevant to the choice of methods. These are differences such as artificial or natural collection context, created or found data, subjective or objective data, and many more. These are the dimensions that distinguish different data collection methods from each other, and they are also the ones that have a major impact on how to analyse and interpret data afterwards. But these divisions, which are also familiar from single method research, also play a different role in mixed methods research. In studies that combine data collection methods, it makes sense to actively use these divisions as tools to think about which methods to combine in your study. It is often these divisions that are bridged by mixing data collection methods.

Many of these differences are well known from the mainstream methodological literature, so in the next section we will highlight two other fundamental differences that are important for the choice of data and data collection methods. They are about the properties of data and are productive starting points for understanding the possibilities, limitations and complementary properties of data. By recognising these fundamental differences, it also becomes clearer why it is often a good idea to work with multiple data types, and what properties they need to have in order to provide evidence for the theories being tested or developed in the study.

Data properties: Opportunities and limitations

Two key distinctions in data properties can be very helpful when developing a mixed methods study:

1. *Narrative vs. action data* relates to what it is about the social phenomenon that our data is meant to gather information about.

2. *Open vs. closed data* relates to what analytical purpose our data should be used for: testing and/or developing theory.

The reason we treat data types here relatively independently of their collection method is that there are many ways to collect data with the same properties. We'll return to collection later in the chapter and link data properties and research styles together again.

Report and action

The first major difference between data is whether data captures narratives or actions. Narrative data captures individuals' statements, typically in an interview situation. All interview-style research data, such as in-depth interviews and survey data, is narrative data. But you would also consider found data such as diaries and social media posts or some experimental data from e.g. survey experiments as narrative data. Action data captures behaviour directly and is independent of the respondent's narrative. Action data is captured and created either through the researcher's own observations, such as ethnographic fieldwork, or through data found in registers that capture actions, or experiments where interventions aim to change behavioural patterns that are directly observed. This could be social media engagement or state registration of work and income, video surveillance data or GPS localisation of people's physical movements.

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Table 4.3 summarises how the research styles can produce narrative and action data respectively.

	Research style			
	Interview	*Observation*	*Found data*	*Experimentation*
Action data	Interviews only capture action in situations where the action is saying or writing something	Observation methods capture all types of actions that are visible to the observer	Many found data captures actions directly or indirectly. For example, registries, GPS data and banking information	Experiments are in most cases designed to capture changes or differences in behaviour
Story data	Interviews are the epitome of narrative data	The vast majority of observation methods capture narrative data either as naturally occurring in the observed or through conversations with the observed	Many data found captures narrative directly, such as diaries, social media posts and media articles	Survey experiments and experiments involving group interactions and discussions capture narrative data

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Some types of narrative data can be treated as action data. For example, if we are not interested in social media posts - what we say about ourselves and the world around us - but are instead interested in how we interact with others and what actions (posts, likes, etc.) we make on the platform, then this data falls into the category of action data. Similarly, when analysing focus group interviews or transcripts of parliamentary debates, you can count who speaks the most and the longest, and who speaks the least and the shortest, thereby capturing action data related to narratives. It is often much more difficult to go the other way and treat behavioural data as narrative data: just because we can observe a certain behaviour, it doesn't mean we know what motivated that behaviour subjectively for that individual.

Much of social science uses the interview style, either as small-scale in-depth interview data or large-scale questionnaire data, and makes statements about both attitudes and actions based on this (Jarolmack & Khan, 2014). This ignores the fact that people often say one thing but do something else. For example, some people think the climate crisis is a big problem that everyone has a responsibility for, but at the same time they take the plane on holiday a few times a year.

The discussion on the fundamental difference between action data and narrative data illustrates two key points (Jarolmack & Khan, 2014). Firstly, it is clear that some data types are best at capturing attitudes and narrative, while others are best at capturing actions. The criticism is particularly levelled at studies that use narrative data and make assumptions about actions without the necessary caveats, as in the example of climate and aviation. Conversely, much economic research often makes assumptions about people's attitudes and preferences based on their actions.

Secondly, the distinction between narrative data and action data is not determined by what the data collection is about, but how the data is generated: Collecting interviews about actions, such as how respondents voted in the last election, does not mean that the interviews are action data. They are still narrative data, as you have collected accounts of what respondents voted and not observations of the actual action in the voting booth. The same is true in the reverse case: you cannot observe actions and

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claim to have observed narratives. For example, you can collect data on what posts people share on social media, but it cannot tell us whether they shared it because they liked the content or just didn't like the content.

Researchers often choose between narrative data and behavioural data based on how costly it is to obtain one or the other type of data. For many years, if you want to study people's consumption, it has been far easier and cheaper to collect narrative data by asking people how often they shop than to collect action data by monitoring them over a long period of time and counting the number of purchases. With the emergence of digital systems that capture and store behavioural data, such as supermarkets' records of purchases through checkout scans and payment systems, the cost of obtaining behavioural data has fallen dramatically.

The problem with a lot of action data, especially found digital data, is that it can be difficult to interpret because you don't have the actor's own understanding of the action. For example, we can easily register when and what a given person buys, but we don't know why and for what purpose and for whom. Is it personal consumption? Do the purchases cover the consumption of an entire household, or are they for a child's birthday party or a sick relative who lives elsewhere? Pure behavioural data is often quite unhelpful when trying to answer questions that go beyond very simple descriptions.

Many researchers solve this problem with behavioural data by making theoretically informed guesses about people's motives, but with mixed methods research, you can solve the problem better and with stronger evidence: If we added narrative data by, for example, interviewing the buyers, we could say much more and with greater certainty. In order to understand actions, we often need to know the actors' own understanding of the actions, which is why there are many situations where the movement between narrative data and action data is optimal.

This is precisely what motivates many complementary mixed methods designs: using narratives about how people experience and interpret a

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given situation combined with data about how they act in the situation. Attempting to capture both narratives and actions with narrative data alone significantly lowers the quality of the study, while the mixed methods approach ensures the highest quality.

In text box 4.2 we provide an example of a study that manages to combine narrative data with action data to better understand the consequences of certain events.

Box 4.2 The origins and consequences of solidarity highlighted in both action and narrative

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In their study "The Solidary Relationship's Consequences for the Ebb and Flow of Activism: Collaborative Evidence from Life-History Interviews and Social Media Event Analysis", Carlsen and colleagues (2020) wanted to investigate how volunteers and activists who wanted to improve conditions for refugees and asylum seekers were affected by meeting and interacting with refugees. To do so, they combined two data sources that both complemented and confirmed each other. Carlsen and colleagues' starting point was life history interviews, i.e. narrative data in which participants talked about how encounters with certain refugees had been turning points in their careers as activists and volunteers

because they felt an enormous responsibility for those refugees,

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who they saw in a distressed and hopeless position. They felt an ethical obligation to help, which developed into solidarity with the refugees and led to extensive and often long-lasting involvement. This connection seemed plausible at first glance, but it was difficult to determine whether the interviewees would have been active regardless of whether they had met the refugees in question, just with a different justification. This could not be determined with the narrative data collected. To shed light on whether meeting refugees increased activity afterwards, they used action data from Facebook, where thousands of pro-refugee activists were organised into groups where they coordinated help and where refugees were also active. By classifying each profile name according to whether it was likely to refer to a refugee or activist, they were able to examine whether those activists who had an interaction with a refugee, such as a conversation in a thread or a tag, were subsequently more active and for longer than those who did not. The study showed that an interaction between activist and refugee had a positive effect on activity levels in the weeks after the interaction took place. In this way, the behavioural patterns from the action data

confirmed the interpretations and narratives provided by the interviewees in the narrative data from the interviews. Conversely, the narrative data complemented the action data by providing evidence for the experiences and interpretations that could explain the observed behavioural patterns.

Open and closed data

Another important difference between different data properties is the distinction between closed and open data types. While the difference between narrative data and action data was used to illustrate how different data types have strengths and limitations in capturing different parts of the research subject, open and closed data types are important in addressing and combining different types of the research questions we introduced in Chapter 3 (Table 3.1).

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The distinction between open and closed questions is well known in survey research. Open-ended questions are defined by the fact that there are no specific answer options and the respondent can instead formulate the answer in their own words, typically in the form of a piece of text typed into a text box. In contrast, closed questions are what we typically associate with survey questions, where the range of possible answers is given in advance as a scale from 1 to 7 or as Yes/No/Don't know. This distinction can be applied to a wide range of data beyond survey data.

For example, think about the difference between a reaction and a social media post. Reactions are limited to certain emoticons and can be considered closed. A written post, on the other hand, is open because the user must express themselves in their own words to communicate with their network. The open or closed nature of data offers vastly different possibilities in terms of what knowledge we can generate and what questions we can answer.

Table 4.4 summarises how open and closed data can be produced in the four method styles.

Table 4.4 Open and closed data production within the research styles

	Research style			
	Interview	*Observation*	*Found data*	*Experimentation*
Open data	Semi- or unstructured interviews or surveys with fieldwork and other similar passively produced text, produced through social open response options, forms of observation and video, audio and similar interaction experiments, typically text boxes	Open ethnographic participant observation	All forms of found or	Open data can be for example, but it is rare
Closed data	Survey with closed response categories	Structured observations that classify observations into defined systems	All kinds of registers, databases, collections and ar es that use classifications	Most experiments produce closed data

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Closed data forms are effective because they are easy to scale, both in the collection itself and in the analysis. You can calculate a simple average when all respondents have categorised their answers as a value between 1 and 7, whereas it is difficult and scientifically questionable to create an average over a series of in-depth, open-ended interviews. It is also easier to collect and analyse large amounts of standardised survey responses than it is to conduct many in-depth interviews where respondents can answer in their own words and influence what to talk about. The standardised form of closed data therefore makes it easier to compare the phenomena it captures across units of analysis.

These properties make closed data a strength when answering a research question of the theory-testing type in Table 3.1, because when we need to test a statement, we need comparable data. If we have a hypothesis that political activists who have experienced being treated badly by the police lose trust in the Danish Parliament, the judiciary and the police, we need a comparable measure of the activists' possible loss of trust in the three institutions. With closed data from a questionnaire about whether the respondent has participated in political activism and whether they have had positive or negative experiences with the police, we will be able to easily distinguish between those who have been exposed to the influence and those who have not. We can then compare their answers to questions about loss of trust in parliament, the judiciary and the police. This ensures comparable data so we can test our hypothesis by counting which institution most people ticked that they had lost trust in and whether there is a difference between activists with positive and negative experiences with the police. This doesn't mean that closed data can only be used for testing - far from it - and it certainly doesn't mean that open data can't be processed in a way that can answer a testing research question. But the closed data types are easier to use for testing purposes, and they can often be collected on a large scale, enabling theory-testing forms of analysis that cannot be applied to smaller-scale data.

Closed data is often more limiting when we ask questions about something we don't know much about, i.e. when we ask questions of the theory-developing type in table 3.1. We are often unsure of what

exactly characterises what we want to investigate, what is relevant and important to understand about a given phenomenon, and perhaps also the boundaries of the phenomenon itself. Open data types are more suitable here because they allow the actors to freely describe what they are doing without being subject to the closed data types' restrictions on what can be said or answered. Closed data types assume that you know what the relevant answers might be, whereas open data formats allow those you are analysing to help define what the right questions are. This also applies to observation, where open observation in fieldwork, for example, leaves the researcher free to pursue and explore whatever seems significant in the situation (Harper, 1992).

In the example of activism and experiences with the police, it would be important to understand which experiences with the police affect trust and why those experiences affect the activist so much. For this purpose, fieldwork and open interviews would be obvious methods. This type of open data would make it possible to identify different types of experiences with institutional actors and activists' interpretations and descriptions of these experiences in relation to their trust in the police, the justice system and the political system. Based on this data, we would be able to create evidence for a theory about the relationship between activists' experiences and their trust in certain institutions, which we could then test with other types of methods and data.

As with narrative and action data, a combination of open and closed data types often makes sense as a motive for a mixed methods project. The combination of open and closed data works both if you have a complementary or an affirmative purpose. A purely complementary purpose is to provide extra information that improves our interpretation of the data, but where the data does not challenge each other's validity. A confirmatory purpose, on the other hand, aims directly at using one data type to assess the quality of another data type.

There are many other key distinctions regarding the nature of data, but these two are essential in most types of research in the social sciences and humanities, where a mixed methods approach has much to contribute.

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The next topic in this chapter is how we can best utilise multiple data sets to create complementary perspectives on the phenomena we are investigating, and thus provide better answers to our overall research questions. The key is data integration.

Data integration

Mixed methods research can combine different data in many different ways, but there is one particular way of combining data that can particularly strengthen a mixed methods study: data integration. Data integration is about how to establish a concrete link between data sets. Such integration is not necessarily always needed in a mixed methods project, but it is important to consider whether data integration is needed to create the right evidence to answer the research question.

In the following, we will first go through the general principle of data integration and then two commonly used types of data integration.

1. *Data integration* generally refers to having multiple data sets relating to the same unit of observation. This could be, for example, interviews with people about how important wealth is to them, combined with data from the tax authorities about the interviewees' wealth, property and income.
2. *Transforming* one type of data into another so that you have two sets of data is a classic technique in quantitative text analysis, for example. Here you often have text data that you transform into numerical, categorised data of different words and types of content.
3. *Nesting* means utilising the hierarchical structure of social reality, enabling a few units of observation at one level (e.g. schools in a municipality) to contain many units of observation at another level (e.g. all classes in the schools and all students in each class).

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Data integration: Multiple datasets about the same unit of observation

As mentioned, data integration means that our datasets are linked by the units of observation being present in all datasets and can be identified in all datasets as illustrated in Figure 4.1. Our unit of observation can be individuals, organisations, places, countries or something else entirely - it depends on the topic of our study. For data integration to occur, there needs to be something that uniquely identifies our unit of observation in all datasets. This could be a name or ID number that allows us to find our in-depth interview data, survey data and observation data on the same person, for example, and compare information from the three data sets.

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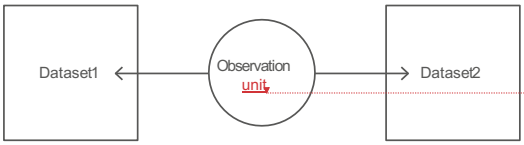

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Figure 4.1 Data integration through observation units

An example that shows the difference between integrated and unintegrated data is the type of field experiments often referred to as interventions. You set up an experiment to see if you can get pupils to read more by doing more reading aloud in primary school. A group of school classes are selected to receive the intervention - reading aloud - and a group of comparable control classes to receive no intervention. Children and parents are asked to answer a questionnaire about reading habits before, during and after the intervention, and fieldwork is carried out in the classes to see how pupils react to the intervention and what impact the teachers' efforts and attitudes as readers have on children's interest. When the intervention is over, we know from fieldwork in the classes that there was a big difference in the teachers' efforts and that it clearly affected the students' motivation.

If the data is completely anonymised and thus not integrated, we cannot identify the different teachers and classes from the fieldwork in the survey data, and therefore we cannot see if the teachers' different efforts had an impact on children's reading outside of school. Instead, if the data is integrated so that we know which class and teacher each student belongs to, we can use the fieldwork data to look at the correlation between each teacher's approach and how it affected children's outcomes. The analytical potential increases dramatically if data is integrated compared to if data is not integrated. Of course, this doesn't mean that anonymity is a big deal, but there may be ways to ensure anonymity that do not jeopardise the analytical benefit.

One of the most common forms of data integration is when respondents who participate in a survey also participate in an in-depth interview. This combination of data can have many purposes, but basically you have one data source that provides large-scale closed narrative data (survey) and another that provides access to small-scale *open narrative data* from a smaller number of respondents from the survey sample (in-depth interviews). As mentioned, data integration is not necessary for creating complementary relationships between data, but it creates some unique opportunities and provides stronger evidence. When different types of data are not on the same unit of observation, there is always uncertainty about whether what the different data sets have captured is really the same and how they relate to each other. If you get some surprising, complementary results, are they really valid or are they simply a result of the two datasets being collected in different locations or with different participants? This suspicion disappears when the data is integrated via the same unit of observation and the relationship between the data sets becomes tangible and concrete.

However, integrating on *all* observation units is often impractical. Often you will have an unequal number of observations in the two datasets, which makes it difficult to compare certain types of results. You can do the read-aloud intervention in many classes over a long period of time, but you can hardly do observational studies in all classes every day. Similarly, it is possible to collect 2000 survey responses, but

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practically impossible to conduct 2000 in-depth interviews. Instead, you might select 20 representative respondents out of the 2000 and conduct interviews with them. Therefore, there will almost always be a mismatch in size between the data sets, and it is often the case that closed data can be collected on a large scale and open data can be collected on a small scale. Even if the open data can be collected on a large scale, it's more demanding in terms of analysing it, so you don't want to work with it on a full scale.

Of the many different choices and options available when designing and conducting a mixed methods study, data integration is something that is highly beneficial to the effort because integrating data enhances the analytical potential of the study and strengthens the evidence that can be mobilised. But of course, this assumes that data integration is possible and meaningful.

In addition to doing data integration on observation units by collecting multiple datasets on the same unit, there are other types of data integration. This can be through creating multiple types of data through so-called data transformation or by utilising the fact that data is often embedded in a group or hierarchical structure that allows the different scales of datasets to be linked to these groups or hierarchical levels. These two particular forms of data integration are discussed below.

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Data transformation

Data transformation means turning one data type into another data type. Many datasets in the digital age have the characteristic of being large and open. Often you can analyse a selection using the techniques we normally use to analyse open data, such as interviews, text or observation, but this often limits how much of the data you can manage to analyse. To utilise large-scale data for statistical analysis, you need to transform the open unstructured text into closed numerical data. Examples of large open data sets could be Facebook, newspaper articles, historical archives or an organisation's internal email communications.

The transformation involves making statistical variables out of the qualitative text material by categorising the data. A typical way to do this is to simply read each text and classify it according to a number of predetermined parameters. This could be the number of words in the text, the language of the text or something more substantial. If we looked at public debate in newspapers, we could classify opinion pieces according to the topics they addressed, whether they were critical, neutral or positive towards the current government, etc. This approach has been used extensively, but it is costly as it requires a lot of resources to read through thousands of texts and classify them. However, digitisation of text and the development of machine learning has made it possible to automate some parts of this process, thus lowering costs significantly.

The result of a data transformation is that you have multiple datasets for all the observation units, i.e. the texts. One dataset is the open text data. The other dataset is closed datasets consisting of variables that capture certain narrowly defined characteristics of the text. Thus, we have complete data integration across all observation units and can analyse both the open and closed data for all units. For example, we can use statistical methods to find patterns based on our variable that, for example, lead to a division of the texts into different groups. Subsequently, by analysing a sample of the open data, we can gain insight into more complex meaning structures that characterise the different groups.

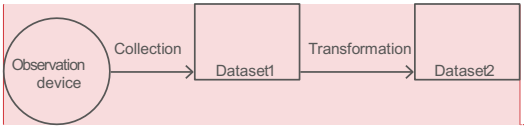

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Figure 4.2 Integration through data transformation

In mixed methods research, this type of transformation is called quantification. The opposite, turning closed data into open data, is called qualification. Quantification is by far the most common data transformation in mixed methods projects. This is because the data produced in a transformation process is quite useful for analysis because you have very complex, open data and transform it into less complex, closed data where you still have enough information to perform an analysis.

In the opposite transformation process - qualification - you start with a closed dataset with limited complexity. Just because you transform it into open data, you cannot add complexity that is not present in the dataset to begin with. Therefore, it is primarily very large or complex closed datasets that can support being transformed into open data.

If you need to use register data to investigate, for example, the impact of psychiatric disorders on children's lives and development, you can use registers to find the people who received the relevant diagnoses as children and then gather all relevant information from their pathways in the healthcare system, in the school and daycare system, with the police and SSP and with the social authorities. For those who have become adults, you can find information about their employment, income, whether they have started a family and any children. This primarily closed data can be analysed using various advanced statistical techniques and provide a picture of how the group of children with psychiatric diagnoses fares later in life. However, you can also perform a data transformation where you select relevant individuals in the data and describe how they have fared by mapping their course over time across registers. A kind of biographical sketch is created for some anonymised, representative individuals that can be used to understand how the statistical model is connected to the sequences of a concrete life, and especially to convey how the findings and correlations in the statistical analysis play out in practice. This is what we call illustration in chapter 5 on analyses.

There are three important considerations to keep in mind when transforming data. Firstly, you need to determine whether your open data is fundamentally suitable for quantification. In-depth interview data is often neither sampled nor generated in such a way that it is comparable across your observations, so quantification makes little sense.

Next, consider whether the categories or themes you are interested in are quantifiable at all. If the category of interest requires a lot of contextual knowledge and interpretive work to identify in the text, it is not suitable for computer-assisted quantification, and even with the use of human classification, those performing the task must be thoroughly instructed and trained to ensure reliable measures.

Finally, consider whether you have the resources required. It takes a long time to manually classify enough text to support a statistical analysis. In transformation, there is often a trade-off between the quality of the classification and the size of the transformed dataset. On the other hand, the technology and methods for computer-assisted analysis of text, video and audio are developing rapidly, so the possibilities are constantly expanding.

Embedded data

Much of the social world is embedded in the sense that parts of social life take place in more or less bounded units. Children go to certain schools and are embedded in them. Volunteers are active and thus embedded in certain organisations or groups. Individuals live and go about their daily lives in certain neighbourhoods and can thus be said to be embedded in the neighbourhood. Neighbourhoods are in turn embedded in districts, which are embedded in postal districts, which are embedded in municipalities, which are embedded in regions, which are embedded in countries, etc. In other words, there is a relatively natural hierarchical structure to many phenomena in the social world, where many entities at one level (e.g. children) are contained within fewer entities at another level (e.g. school classes).

If you have data that tells you which entities at one level are nested within certain entities at a higher level, you have a form of data

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integration called nested data (the principle is illustrated in Figure 4.3). This form of data integration is ideal for a mixed methods study because the results of analyses of the higher-level entities can be used as data about the context that the entities are embedded in, at the lower level - for example, the impact of the size and gender composition of each class on individual student learning. Conversely, analyses of the lower-level units can be used to characterise the higher-level units. For example, we can compare class size and gender composition of classes between schools in the same municipality. Another example is that knowledge of the dominant religious beliefs among residents of different geographical regions in Europe can be used to analyse behaviour or attitudes among the inhabitants of these regions to see if religious culture explains differences among individuals.

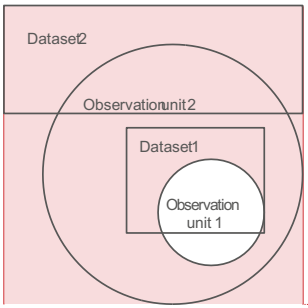

Figure 4.3. Embedded data integration

When working with nested data, you need to make sure that the link between the group level and the individual level is theoretically sound. Is it defensible to claim that the group level influences the individual level? Is the group level actually a context that individuals find themselves in? Such questions are crucial to think through.

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One example could be companies as a context for individuals' job satisfaction. Here it is important to think through what exactly a company constitutes as a context in different cases. For example, some companies are huge and consist of many different workplaces. In this case, the company hardly makes sense as a decisive context for job satisfaction. The workplace is more likely to be the relevant context. In other cases, there is a 1:1 relationship between the workplace and the company, such as in a small craft business or shop.

Embedded data comes in many forms and at different scales. For example, in the case of students, classes and schools, data is available at all levels at large scale in public registers. But embedded data can also be at medium and small scale levels, such as an integrated data design consisting of five small companies (small scale) with a maximum of ten employees at each location, which will yield a maximum of 50 employees (medium scale) at the lower level. It is an important point that although embedded data and, for that matter, data transformation is often associated with large-scale data and statistical analysis, it is not a given and there are good reasons to also benefit from embedding at the smaller scale levels in mixed methods studies.

It is important to point out that data integration can be ethically challenging, both when we are talking about integrating multiple data sources for the same unit of observation and about embedding. When using data integration, it often means that information and observation units exist in several different data sets, regardless of whether they can be directly linked or not. This places special demands on the researcher's ethics and use of anonymisation, not only being aware of the need to make people anonymous in the individual datasets, but also being clear about what data you are presenting across datasets and analyses. It is always the total amount of information that needs to be assessed in relation to anonymity.

Chapter
4

Data collection integration: Sequential and concurrent designs

The mixed methods literature distinguishes between sequential data collection and concurrent data collection (see table 3.2). In sequential data collection, you collect one dataset at a time, typically in a sequence of sub-studies, so you first collect data for sub-study 1, then sub-study 2, etc. This has the key advantage that you can use information from the first data set collected to design the subsequent data collection. In this way, you can optimise complementarity so that you can elaborate on results from the first sub-study or explore new questions that arose from the first sub-study in the second sub-study. This creates a dependency between the different data collections, which can be called *data collection integration*, where knowledge from data collected for study 1 influences the design of the data collection for study 2. The data collection for sub-study 2 is therefore dependent or conditional on the data from sub-study 1, and therefore the data from sub-study 2 cannot be left alone - the data set is only meaningful in the light of the knowledge from sub-study 1 that forms the basis for it.

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Sequential data collection

In sequential data collection designs, you should be clear about the type of dependencies between the different data sets and thus what needs to be done in the first data collection part before the next collection can begin, as illustrated in Figure 4.4. Data collection is often a bottleneck in research projects and especially in mixed methods projects, where there are often at least two bottlenecks. If these are integrated, it requires a well thought-out data collection plan.

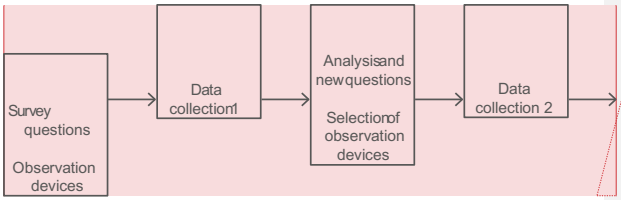

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Figure 4.4 Sequential data collection

We can distinguish between two types of dependency in data collection: *selection dependency* and *analytical dependency*. Selection dependency is when the selection of observation units in the second data collection is dependent on information from the first. Analytical dependency is when the data collection in substudy 2 can only start after analysing the data in substudy 1.

Selection bias can be found in studies where a sequential design is used to create data integration through the units of observation, as we have described earlier in this chapter. An example of selection dependence can be found in Andersen, Toubøl, Kirkegaard and Carlsen's study of informal voluntary help during the COVID-19 epidemic in 2020-2022. Here, a representative survey was first conducted, to be followed up by open-ended interviews. Data from the survey was then used to purposefully select respondents for follow-up in-depth interviews with respondents with certain characteristics of particular analytical interest (Andersen et al., 2022).

An example of analytical dependency can be found in Mario Luis Small's "Someone to talk to" (2017), which examines networks of people who confide in each other and seek emotional support in these relationships. Small is based on an in-depth interview study. On the basis of this, he develops hypotheses about who, when and with what motivations people ask others for emotional support. These results then form the basis for a representative survey, where both the questions and hypotheses are based on knowledge from the previous qualitative analysis. Small moves from theory development in sub-study 1 to theory testing in sub-study 2 in a sequential design.

An important point is that analytic dependence does not necessarily mean that there is selection dependence (and vice versa). For Small's approach in "Someone to Talk to", it was not crucial that the respondents from the in-depth interviews were also part of the representative survey. Small's study is very typical of sequential studies,

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where closed large-scale data used to test theories are analytically dependent on theory-developing analyses of open small-scale data.

Before turning to concurrent data design, it is important to realise that combinations of data types are of course not limited to datasets at different scale levels of interview data as in the examples above. You can combine across interviews, observations, found data and experiments, as the examples elsewhere in the book illustrate. The only limit is your imagination (and resources).

Concurrent data collection

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Sequential data collection is not always desirable, and for students who often have very limited time, it can be a decidedly bad idea. For some research questions, the order of data collection is irrelevant, and in other contexts it is not practical to let one data collection wait for another. Finally, it may be necessary to collect different types of data continuously and in line with the rhythm of social life. This is characteristic of many field studies, for example, where you simultaneously create different data sets with interviews, observations, found data, etc. that together create a basis for understanding a given field. In these cases, simultaneous data collection is preferable.

Concurrent data collection means collecting the data simultaneously and independently of each other as illustrated in Figure 4.5.

Again, if we take the Marienthal study, which we reviewed in Chapter 2, much of the data collection happened simultaneously. Much of the data collection was not dependent on the order in which the data was collected and was therefore not collection-integrated. One way to think about concurrent designs is that the different data covers different areas or different depths, but are basically not dependent on each other in the data collection process itself.

Concurrent data collection can be integrated, but primarily in the form of selection dependency, as analytical dependency requires a sequential order. In the Marienthal study, there was selection dependence because it was the same city, the same families and individuals that all the data collection concerned, and since addresses and names were collected in all datasets, there was also data

integration both via the observation units and embedding. However, there can easily be an iterative process with analytical dependency, where you continuously analyse the individual data sets and use that knowledge to update the focus and questions in the other sub-studies. We will return to this in chapter 5 on analytical designs.

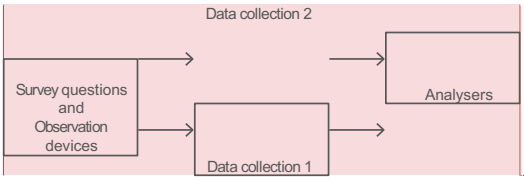

Figure 4.5 Concurrent data collection

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Data quality

Data quality should be assessed in relation to the evidence you want to produce to support, refute or elaborate on the ideas that the study aims to illuminate. In mixed methods research, data quality is also about the specific links between both the datasets and the different sub-studies. Within the different methods, researchers have formulated clear quality criteria for text analysis, experiments and ethnographic fieldwork, for example. When it comes to the data in each sub-study, the quality must of course be assessed against these criteria, because in a mixed methods study the commonly accepted quality standards for interview, observation, found data and experiment still apply, of course. There is a huge literature on this, which we won't go into here (read Martin, 2017, for a good overview).

On the other hand, in mixed methods studies, we add a layer that applies to the relationship between the data sets. In this concluding section, we will present some general considerations about the data quality criteria that arise from the mixed methods approach. Firstly, we will start with some considerations on the importance of *the quantity of data* for quality and how this should be understood across very diverse datasets. We then turn to the specific elements of data quality that are important for mixed methods projects, namely how well the datasets *complement* each other by compensating for each other's weaknesses, how well they *integrate*, and finally how *compatible* they are.

Enough data: Number of observations and degree of exposure

Large amounts of data are typically talked about as the stuff of quantitative studies, but many small-scale qualitative datasets are in fact huge. A normal ethnographic study can consist of thousands of pages of field notes, hundreds of pages of interview transcripts and huge amounts of archive material, images, etc. The size of a dataset

should not be confused with the number of observation units. But a crucial quality criterion is that you have enough data to be able to derive strong evidence for your conclusions.

Statistical researchers generally believe that a large number of observations, i.e. large-scale data, is a good thing. Large-scale data makes it possible to be more certain whether your results are driven by chance and to investigate differences and rare events in your data (Salganik, 2018). These characteristics are important for statistical studies, but also provide better opportunities for integration in mixed methods projects. Many observations make it possible to not only look at the average across the entire population, but also explore subgroups within the population. This gives you a better starting point for integrating with open small-scale data because you focus on specific subgroups that you can explore in depth. The quality of a mixed methods data integration across sub-studies will often partly depend on having enough observation units in your large-scale dataset.

For example, in Andersen, Toubøl, Kirkegaard and Carlsen's study of informal voluntary help during the COVID-19 epidemic in 2020-2022 (Andersen et al., 2022), the researchers were interested in collecting indepth interviews with the very hard-to-reach group of highly socially vulnerable citizens who are generally not easy to get to volunteer for interviews. However, a survey sampling frame with a relatively large number of respondents enabled them to identify socially vulnerable citizens. Subsequently, they could contact and interview them, as they had asked in the survey if they could contact the participants for a possible follow-up interview. Here, the large number of survey respondents was crucial to the quality of the strategic selection of interview participants.

For researchers working with small-scale open data, the size of the data is about more than the number of units of observation. Without saying that the number of units doesn't matter, they are at least as concerned with the amount of exposure to the phenomenon they are studying. The size and quality of an ethnographic dataset is generally

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not determined by how many people you have spoken to during your fieldwork, but by how long the fieldwork has been going on and how far you have travelled into the field. In open-ended, in-depth interviews, the length of the interviews, and thus the amount of exposure in the interview situation, is an essential criterion. Exposure is an essential criterion for quality in open data collection, but not the only one.

Quality depends on many things, depending on the type of data, and here you have to turn to the specialised methodology books. But the most typical understandings of quality related to exposure concern whether the interviewee has been given the necessary trust to gain credible knowledge, whether they have been exposed enough to develop a sufficient and credible understanding of the field, or whether they have read enough text material to gain a sufficient understanding of the variations and content of the overall material. All of these factors are crucial to the credibility and thus the quality of the data when working with open data. Very superficial open-ended interviews run the risk of becoming unstructured surveys with neither breadth nor depth, which can neither confirm nor supplement the results of other sub-studies in a credible way. Similarly, there are many examples of anthropologists who do not achieve the understandings and insights that lead to epistemological breakthroughs until well into their fieldwork (Timmerman & Tarvory, 2014).

The threats to data validity

All data has different strengths and weaknesses, and the differences are related to whether it is open or closed data, action or narrative data, or whether it is large or small scale data. One of the ways mixed methods studies can increase the quality of a study is by combining data types that complement each other's weaknesses or blind spots. One of the key motivations for mixed methods research - particularly for confirmatory designs - is thus in fact, trying to utilise the strengths of multiple types of data to overcome some of the weaknesses that mono-method designs suffer from. For example, it's very difficult to make sure

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that a survey question captures a particular action correctly: even if you try to capture it by asking in five different ways, you can't avoid the fact that narrative data suffers from some weaknesses when it comes to capturing actions. Therefore, in cases where you want to investigate actions and behaviours, the combination of survey and in-depth interviews is not the best possible combination of data, because both produce narrative data on different scales.

Each specific data set has different strengths and weaknesses in relation to the evidence you can produce, which is why it's about combining data that together best covers the weaknesses of the individual data in relation to the theory you need to create evidence for. Surveys are an effective way to say something about many people, but they are an uncertain way to capture the behaviour of many people. If you are interested in behaviour, a supplementary dataset with behavioural data will optimally shed light on whether the survey captures behaviour correctly or not. For example, if we are interested in the climate impact of a group of people's consumption, we could collect data on many people with a survey and supplement the survey with a daily count of a smaller group's purchases or by mapping their waste to get action data that can complement the survey's narrative data.

One way to get an overview of which data combinations would be the best is to start with the research questions and consider what type of evidence they require. Then try to find the combinations of data sources that have these qualities in common.

The power of the integration

In general, a strong integration of data is considered to increase the quality of a mixed methods study. This is because the complementarity between sub-studies and datasets becomes more tangible and can directly verify how what you observe in one type of data relates to what you observe in another type of data. Thus, we can assess this quality criterion through the integration principles we have discussed above.

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Data integration increases quality in terms of credibility, transparency, validity and reliability because it allows you to follow the connections between the information in all datasets and relate them concretely to each other, allowing you to analyse, re-analyse and test in light of the doubts that always arise during the study. However, there is one notable exception. If you're designing a confirmatory study, it's the other way round: Here, collection and data integration is clearly a weakness, as it violates the principle of the independence of the two data sets and thus increases the risk that both a confirmation and rejection of the hypothesis is due to mutual influence between the data.

Compatibility

For certain questions, it is important that the different types of data refer to the same phenomenon, i.e. that they can capture the same thing. The question here is whether an idea developed with one type of data can be investigated with the other type of data, or whether the collection methodologies for the different data sets distort the original idea too much for it to be said that the two data types capture the same phenomenon. A common criticism of mixed methods studies is that researchers think they are investigating the same thing in the substudies, but this is not the case as their two data sets and analyses have actually led to the construction of two different, related phenomena.

You should therefore ask yourself whether the knowledge and understanding gained from in-depth interviews, for example, can be elucidated through actionable data on a large scale? Can the in-depth analysis of text be quantified in a meaningful way? Scaling data often requires closed data where large amounts of information are lost - from a half-hour interview on a topic to a survey question with a response scale from 1 to 5 - so it's important to make yourself and the reader aware of how observations in one dataset differ from observations in the other in order to assess quality.

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The key point here is not only the fact that at higher scaling levels, the amount of information for each unit of observation is typically much smaller and thus very different from the information at lower scaling levels. The critical question regarding the quality of the conclusions drawn from the data is whether what is captured with large-scale data can be said to be the same phenomenon or reflect key characteristics of the phenomenon captured with small-scale data.

Summary

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In this chapter, we have argued that collecting multiple types of data enables more creative and effective use of data and supports more robust evidence for theories. This is because different data illuminates different aspects of social reality. Sticking to one type of data limits both the theories you can develop and the questions you can investigate. In this chapter, we have also reviewed different data collection methods and data types, their strengths and weaknesses, and when to combine them. The latter is partly determined by the theories you have about your phenomenon and partly by the analytical purpose of the study. We then looked at data and data collection integration, which relates to how data from different data sets are linked to each other and how different data collections are related. We distinguished between different ways in which data can be integrated, listed their advantages and disadvantages, and described sequential and simultaneous data collection designs. Finally, we have reviewed different data quality criteria related to mixed methods studies.

Recommended literature:

For more on the relationship between theory and data design and specific guidance on how to align data design and theory:

- Abbott, A. D. (2004). *Methods of Discovery: Heuristics for the Social Sciences (Contemporary Societies*). WW Norton & Company.

A good introduction to different ways of thinking through the relationship between theory, phenomenon and data collection:

- Martin, J. L. (2017). *Thinking through Methods: A Social Science Primer*.
  University of Chicago Press.

For more on the possibilities of digital data and its combination with conventional data:

- Salganik, M. J. (2019). *Bit by Bit: Social Research in the Digital Age*. Princeton University Press.